SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science ASSESSMENT TOOL 1 – Data Interpretation

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Course Code:	DSA03	**Course Title:**	Natural Language Processing
CO Assessed:	CO4 – Precision	**Assessment:**	Assessment Tool 1 – Data Interpretation
Weightage:	40% of CIA		
CO5 Statement:	Formulate context-free grammars, feature structures, probabilistic parsing techniques and to construct parsing frameworks. (BL-4)		
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Section / Batch:	CSE AI	**Date:**	11/08/26

Code of Conduct

*I **A. Mounish / 192472273** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: ___________________________

Instructions

- Read each data set, table, semantic representation, or scenario carefully before answering.
- Analyze the given information and provide logical, evidence-based interpretations.
- Show all intermediate reasoning, semantic analysis steps, predicate representations, or calculations wherever applicable.
- Answers should be concise, structured, and technically accurate.
- Support your conclusions using the data provided in the question.
- Draw diagrams, semantic networks, parse trees, or logical representations wherever relevant.
- Avoid copying definitions alone; focus on analyzing the given scenario and interpreting the data.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.

Representing Meaning & Meaning Structure of Language	Analyze semantic representations, identify action–entity relationships, detect interpretation errors, and evaluate meaning representation in chatbot systems.	Q1
First-Order Predicate Calculus & Representing Linguistically Relevant Concepts	Construct predicate representations, apply logical inference rules, derive conclusions, and evaluate reasoning in smart manufacturing environments.	Q2
Lexemes and Their Senses & Word Sense Disambiguation	Analyze contextual clues, determine correct word senses, evaluate ambiguity resolution, and improve semantic understanding in search systems.	Q3
Syntax-Driven Semantic Analysis	Analyze syntactic structures, assign semantic roles, evaluate parsing accuracy, and improve semantic interpretation in healthcare NLP applications.	Q4

Annexure A – Data Interpretation

1. Semantic Representation in Customer Support Chatbots.

A telecom company collected customer chatbot interactions. The semantic analyzer generated the following meaning representations.

Customer Query	Semantic Representation
"Activate international roaming for my number."	ACTIVATE(Roaming, Customer)
"Deactivate caller tune service."	DEACTIVATE(CallerTune, Customer)
"Check my data balance."	QUERY(DataBalance, Customer)
"Enable 5G service."	ACTIVATE(5GService, Customer)

Additional chatbot logs show:

Query ID	Actual User Intent	Predicted Intent
Q1	Activate Roaming	Activate Roaming
Q2	Deactivate Caller Tune	Activate Caller Tune
Q3	Check Data Balance	Query Data Balance
Q4	Enable 5G Service	Activate 5G Service

Tasks:

1. Analyze the semantic representations and identify the action-object relationships.

2. Determine which query contains semantic interpretation errors.
3. Evaluate how meaning representation affects chatbot decision accuracy.
4. Recommend improvements to enhance semantic understanding in telecom customer support systems.

2. First-Order Predicate Calculus for Smart Manufacturing.

An Industry 4.0 manufacturing plant monitors machines using semantic rules.

Production Data:

Machine	**Status**
M1	Active
M2	Active
M3	Maintenance
M4	Active

Predicate Rules:

- Active(x) → Producing(x)
- Produces(x,y)∧ Active(x) → Available(y)
- Maintenance(x) → ¬Producing(x)

Tasks:

1. Represent the production data using First-Order Predicate Calculus.
2. Analyze which products are currently available.
3. Infer whether Gear production is affected by maintenance activities.
4. Evaluate the effectiveness of predicate logic in industrial decision-support systems.

3. Word Sense Disambiguation in E-Commerce Search Engines. An e-commerce company observed the following search logs.

User Query	**Possible Sense 1**	**Possible Sense 2**
"Apple accessories"	Fruit	Technology Brand
"Mouse wireless"	Animal	Computer Device
"Java tutorial"	Island	Programming Language
"Python course"	Snake	Programming Language

Search Context Data:

Query	**Clicked Result**
Apple accessories	iPhone Charger
Mouse wireless	Bluetooth Mouse
Java tutorial	Coding Lessons
Python course	Software Development Training

Tasks:

1. Analyze the contextual information and determine the correct word sense for each query.
2. Identify the semantic cues used for disambiguation.
3. Evaluate the impact of incorrect sense selection on search engine performance.

4. Suggest an industrial-scale WSD strategy for improving recommendation accuracy.

4. Syntax-Driven Semantic Analysis in Healthcare Information Systems.

A healthcare NLP system processes medical reports.

Parsed Sentences:

Sentence	**Parse Structure**
Doctor prescribed medicine to patient.	Subject-Verb-Object
Patient reported severe headache.	Subject-Verb-Object
Nurse monitored patient continuously.	Subject-Verb-Object
Medicine reduced blood pressure.	Subject-Verb-Object

Semantic Roles Generated:

Entity	**Role**
Doctor	Agent
Medicine	Instrument
Patient	Recipient
Headache	Symptom

Tasks:

1. Analyze how syntactic structures contribute to semantic role assignment.
2. Determine whether the assigned semantic roles are appropriate.
3. Evaluate potential errors that could arise from incorrect parsing.
4. Recommend methods to improve syntax-driven semantic analysis in healthcare applications.

Submission Note
Prepared strictly from the supplied assessment document. Answers are structured to address each task directly, with semantic representations, logical inference, contextual analysis, and diagrams.

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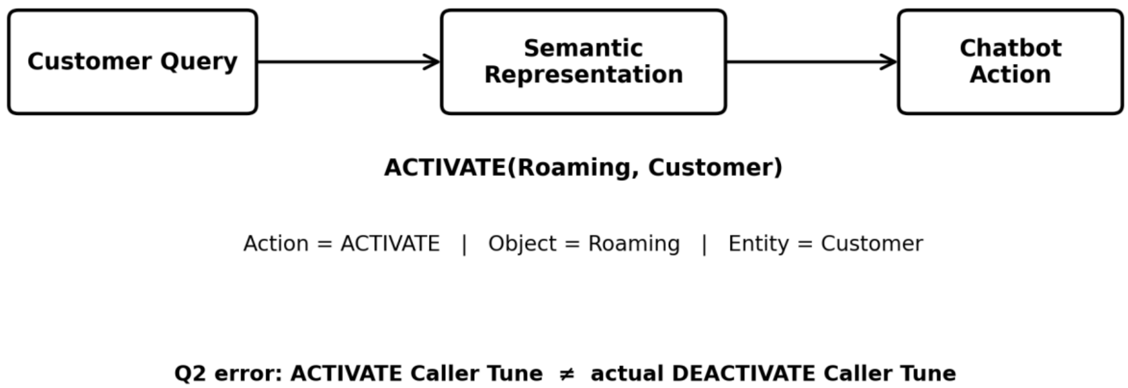
Q1. Semantic Representation in Customer Support Chatbots

The semantic representations use the form ACTION(Object, Entity). They convert each customer request into a structured representation that can be interpreted by the chatbot decision system.



1. Action–Object Relationships

Customer Query	Semantic Representation	Action	Object	Interpretation
Activate international roaming for my number.	ACTIVATE(Roaming, Customer)	ACTIVATE	Roaming	Activate international roaming
Deactivate caller tune service.	DEACTIVATE(CallerTune, Customer)	DEACTIVATE	Caller Tune	Deactivate caller tune
Check my data balance.	QUERY(DataBalance, Customer)	QUERY	Data Balance	Check current balance
Enable 5G service.	ACTIVATE(5GService, Customer)	ACTIVATE	5G Service	Activate 5G service

2. Semantic Interpretation Error

Q2 contains the semantic interpretation error. The actual intent is “Deactivate Caller Tune”, whereas the predicted intent is “Activate Caller Tune”. The object is identified correctly, but the action is reversed.

Correct representation
DEACTIVATE(CallerTune, Customer)

Incorrect predicted action
ACTIVATE(CallerTune, Customer)

3. Effect on Chatbot Decision Accuracy

Meaning representation directly affects the action selected by the chatbot. In the four supplied queries, Q1, Q3 and Q4 are correct while Q2 is incorrect.

Intent accuracy = (3 / 4) × 100 = 75%. Therefore, the sample chatbot achieves 75% intent accuracy, and the Q2 action reversal can cause the system to perform the opposite service operation.

4. Recommended Improvements

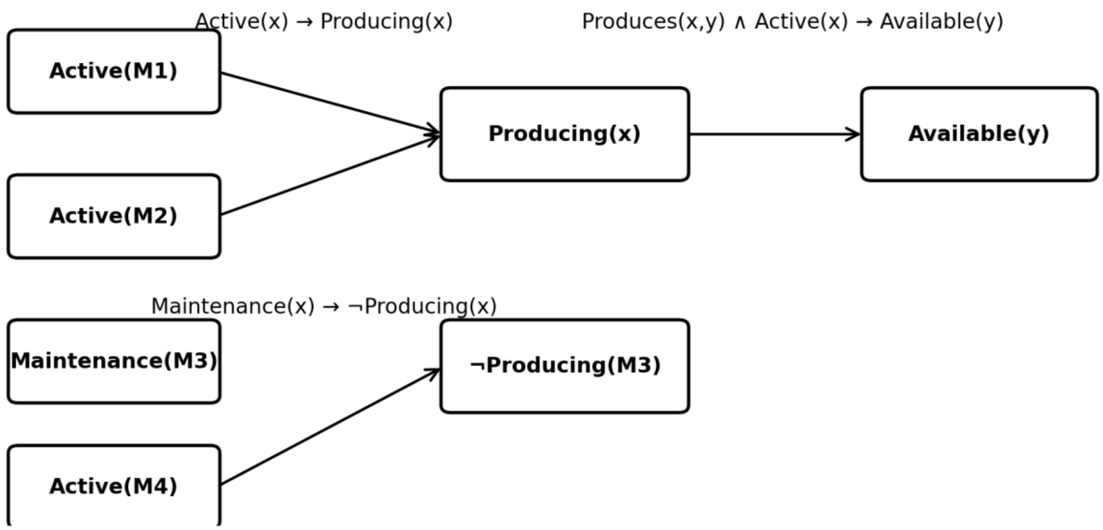
1. Strengthen intent classification so that opposite actions such as activate and deactivate are clearly separated.
2. Use entity recognition to identify telecom services such as roaming, caller tune, data balance and 5G.
3. Apply context analysis to understand the complete customer request.
4. Implement explicit negation and deactivation handling.
5. Use confidence thresholds and confirmation for uncertain predictions.
6. Expand domain-specific telecom training examples.

Conclusion

Accurate semantic parsing is essential because an incorrect action predicate can cause the chatbot to execute an operation opposite to the customer's request.

Q2. First-Order Predicate Calculus for Smart Manufacturing

The supplied manufacturing data states that M1, M2 and M4 are Active, while M3 is under Maintenance. The given predicate rules allow production status and availability to be inferred.



1. Production Data in First-Order Predicate Calculus

Machine facts:

- Active(M1)
- Active(M2)
- Maintenance(M3)
- Active(M4)

Given rules:

- Active(x) → Producing(x)
- Produces(x,y) ∧ Active(x) → Available(y)
- Maintenance(x) → ¬Producing(x)

Applying the rules gives:

- Active(M1) → Producing(M1)
- Active(M2) → Producing(M2)
- Maintenance(M3) → ¬Producing(M3)
- Active(M4) → Producing(M4)

2. Products Currently Available

The rule Produces(x,y) ∧ Active(x) → Available(y) states that a product is available when an active machine produces it. However, the supplied data does not specify any Produces(x,y) facts.

Evidence-based conclusion
Specific product names cannot be determined from the supplied data. Any product produced by M1,

M2 or M4 would satisfy the availability rule.

3. Effect of Maintenance on Gear Production

Because M3 is under maintenance, Maintenance(M3) implies ¬Producing(M3). However, the assessment does not provide the fact Produces(M3,Gear). Therefore, Gear production cannot be conclusively stated to be affected.

Logical conclusion
Gear production is affected only if M3 is responsible for producing Gear. The supplied data does not establish this relationship.

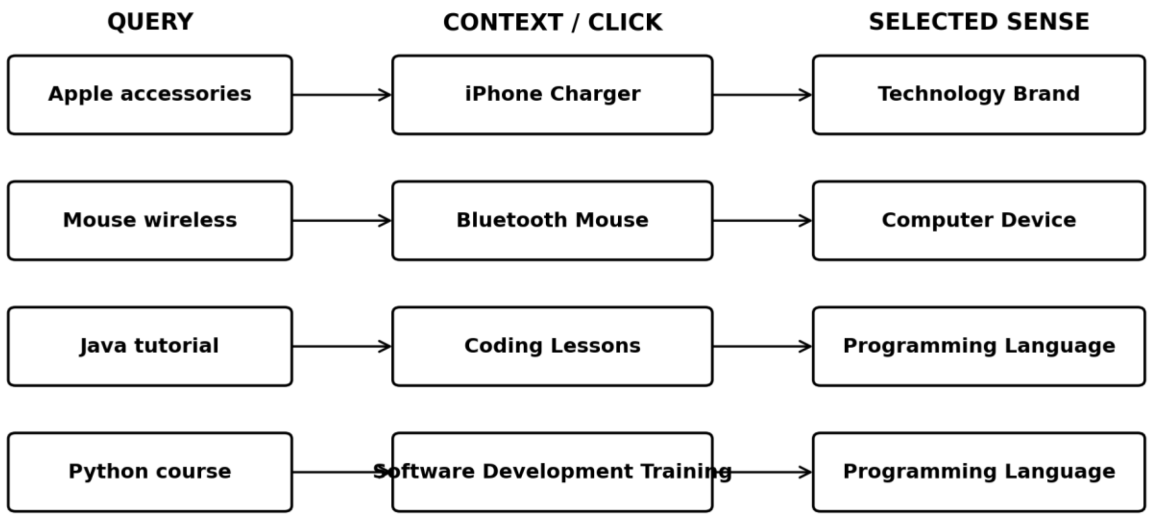
4. Effectiveness of Predicate Logic

- Represents machine states precisely.
- Supports automated logical inference.
- Provides explainable rule-based decisions.
- Can identify production and maintenance conditions.
- Requires complete and accurate production relationships for stronger conclusions.

Conclusion
First-Order Predicate Calculus is effective for explainable industrial decision support, but its conclusions depend on the completeness of the supplied facts.

Q3. Word Sense Disambiguation in E-Commerce Search Engines

The supplied search logs contain ambiguous terms whose intended senses can be inferred from the clicked search results.



1. Correct Word Sense for Each Query

User Query	Clicked Result	Correct Sense	Reason
Apple accessories	iPhone Charger	Technology Brand	iPhone is associated with Apple technology products.
Mouse wireless	Bluetooth Mouse	Computer Device	Bluetooth Mouse is a computer input device.
Java tutorial	Coding Lessons	Programming Language	Coding lessons indicate Java programming.
Python course	Software Development Training	Programming Language	Software development training indicates Python programming.

2. Semantic Cues Used for Disambiguation

- Words surrounding the ambiguous term in the query.
- The clicked result selected by the user.
- Technology and education domain context.
- Product associations such as iPhone Charger and Bluetooth Mouse.
- Programming-related associations such as Coding Lessons and Software Development Training.

3. Impact of Incorrect Sense Selection

Incorrect sense selection can produce irrelevant search results and recommendations. For example, interpreting Apple as fruit, Mouse as an animal, Java as an island, or Python as a snake would move the search away from the user's intended technology or education context.

- Lower search relevance.
- Incorrect recommendations.
- Reduced user satisfaction.
- Poor result ranking and personalization.
- Lower engagement with the search system.

4. Industrial-Scale WSD Strategy

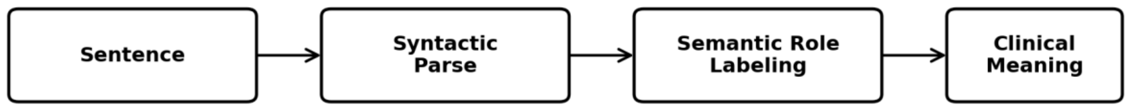
7. Use contextual language models to understand surrounding words.
8. Combine query history and clicked-result behavior.
9. Use knowledge bases to connect terms with structured entities.
10. Maintain domain-specific sense dictionaries.
11. Use personalization based on previous interactions.
12. Apply feedback from clicks, purchases and user behavior.

Conclusion

The supplied logs show that contextual evidence is highly effective for resolving ambiguous terms and improving recommendation accuracy.

Q4. Syntax-Driven Semantic Analysis in Healthcare Information Systems

The healthcare NLP system uses sentence structure to assign semantic roles to medical entities. The supplied data includes four parsed sentences and a semantic-role table.



Doctor → Agent **Medicine → Theme/Object** **Patient → Recipient**

Headache → Symptom | Roles depend on sentence context

Important correction: "Medicine" is not necessarily an Instrument.

1. Contribution of Syntax to Semantic Role Assignment

Syntactic structure identifies the relationship among the subject, verb and object. This structural information provides the basis for semantic role assignment.

> **Example**
> Doctor prescribed medicine to patient → Doctor = Agent; Medicine = Theme/Object; Patient = Recipient.

For "Patient reported severe headache", the patient is the experiencer and headache is the symptom. Therefore, syntax helps connect entities with the action or event described by the sentence.

2. Evaluation of Assigned Semantic Roles

Entity	Assigned Role	Evaluation
Doctor	Agent	Appropriate – performs the prescribing action.
Medicine	Instrument	Needs refinement – in the prescribing sentence it is better treated as the Theme/Object.
Patient	Recipient	Appropriate – receives the medicine.
Headache	Symptom	Appropriate – represents the reported symptom.

3. Potential Errors from Incorrect Parsing

- Incorrect identification of who receives or performs a medical action.
- Incorrect classification of medicine as an instrument instead of the object/theme of prescribing.
- Failure to identify headache as a symptom.
- Incorrect interpretation of the causal role of medicine in reducing blood pressure.
- Errors in medical information extraction and clinical decision support.

4. Methods to Improve Syntax-Driven Semantic Analysis

13. Use dependency parsing to identify grammatical relationships.
14. Apply Semantic Role Labeling for Agent, Patient, Theme, Recipient and Experiencer roles.
15. Use medical-domain language models trained on clinical text.
16. Use medical terminology resources and domain knowledge bases.
17. Apply context-aware parsing across complete clinical sentences.
18. Train and evaluate using accurately annotated medical reports.
19. Add validation rules to detect inconsistent or medically implausible role assignments.

Conclusion

Syntax-driven semantic analysis is valuable for healthcare NLP, but semantic roles must be assigned according to the actual sentence context. In the supplied role table, Medicine → Instrument requires refinement.

OVERALL CONCLUSION

The four assessment scenarios demonstrate how NLP systems represent meaning, perform logical inference, resolve lexical ambiguity and connect syntax with semantic roles.

Question	NLP Concept	Key Finding
Q1	Semantic Representation	Q2 has an Activate/Deactivate interpretation error; sample intent accuracy is 75%.
Q2	First-Order Predicate Calculus	M3 is not producing because it is under maintenance; Gear impact cannot be confirmed without a production relationship.
Q3	Word Sense Disambiguation	Context and clicked results identify the intended senses of Apple, Mouse, Java and Python.
Q4	Syntax-Driven Semantic Analysis	Syntax supports semantic role assignment, but Medicine → Instrument requires refinement.

The answers emphasize evidence-based interpretation, logical reasoning, application of NLP concepts and clear conclusions, matching the assessment's stated expectations.

Student Signature	Faculty In-charge	Course Coordinator
____________________	____________________	____________________